Technical Parameters

Width*depth*height	570mm*680mm*780mm	Weight	130kg
Cooling system	Automatic air-cooling spindle	Motor	AC servo
Number of axis	5	Voltage/Power	220V/2.6KW
Tool type	**Zirconia Milling Bur:** ϕ4mm*(2mm/1mm/0.6mm) Ball-mill, ϕ4mm*1mm End-mill **Resin Millng Bur:** ϕ4mm*(2mm/1mm)		
Automatic tool length detection	✓	Rotation angle	-30°~+40°
Automatic tool life monitoring	✓	Spindle speed	60000RPM
Built-in-burs	6	Fields of application	Dry milling
Environmental temperature	5°C~40°C	Air pressure	＞0.65Mpa
Remote control	✓	Monitor	9.7"LCD touch screen

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Intelligent 5-axis Dental Milling Machine

BSM-520D

5 Axis | Dry Milling | ±0.005 mm | Open | Labside



Intelligent 5-Axis Dental Milling Machine

BSM-520D

CAD/CAM Digital Processing



The BSM-520D 5-axis dental milling machine is engineered for labs demanding uncompromising accuracy without sacrificing efficiency.

Featuring 90° vertical milling and ultra-wide B-axis rotation, it handles intricate details with ease, while the 1/3 holder boosts ROI by 20%—perfect for precision-driven production.



Millable Materials



Zirconia | PMMA | Wax | PEEK

Millable Restorations



Full contour crown | Coping | Inlay/onlay | Veneer | Crown bridge | Zirconia abutment | Zirconia implant bridge | Non-metal frame

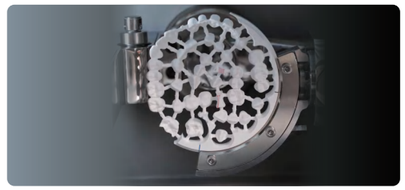
Key Features

High compatibility



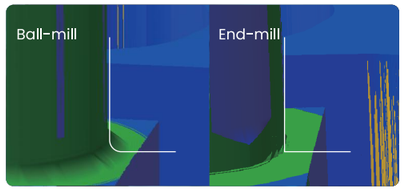
- Wide B-axis rotation (-30° to +40°)
- **90° vertical milling** precisely replicates natural tooth anatomy and surface texture

Higher ROI



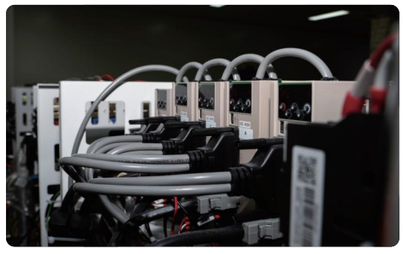
- 1/3 holder boosts output by 20% per block
- **Dust-proof design** extends machine lifespan

Intelligent Processing



- **Automatic changer for 6 tools** with haptic detection & breakage monitoring
- Integrated PC with **9.7 inch intelligent touch screen**
- Configuring a **end milling program** enables better handling of shoulders and corners.

High Accuracy



- **Digital servo system** with high resolution
- Guaranteed milling accuracy: **±0.005mm**
- ±0.01mm installation accuracy per axis

Mlilling speed:

zirconia anterior anatomic crown ≈ 12′

zirconia posterior anatomic crown ≈ 17′;

zirconia 3-unit crown bridge ≈ 30';

The above data was obtained based on the BSM-520D milling of Aconia 3D Master 14mm.

Excellent Reliability



- **60,000 RPM premium spindle** for high-speed milling
- Industrial-grade **aerometal gantry structure** ensures ultimate reliability